

Section 7.3 Exercise Answers

1. $\sin(2x) = \frac{3\sqrt{7}}{32}, \cos(2x) = \frac{31}{32}, \tan(2x) = \frac{3\sqrt{7}}{31}$
2. $\sin(2x) = \frac{4\sqrt{5}}{9}, \cos(2x) = -\frac{1}{9}, \tan(2x) = -4\sqrt{5}$
3. $\sin(2x) = \frac{\sqrt{3}}{2}, \cos(2x) = -\frac{1}{2}, \tan(2x) = -\sqrt{3}$
4. $\sin(2x) = -\frac{16}{65}, \cos(2x) = -\frac{63}{65}, \tan(2x) = \frac{16}{63}$
5. $\cos \theta = -\frac{2\sqrt{5}}{5}, \sin \theta = \frac{\sqrt{5}}{5}, \tan \theta = -\frac{1}{2}, \csc \theta = \sqrt{5}, \sec \theta = -\frac{\sqrt{5}}{2}, \cot \theta = -2$
6. $\cos \theta = -\frac{\sqrt{2+\sqrt{2}}}{2}, \sin \theta = -\frac{\sqrt{2-\sqrt{2}}}{2}, \tan \theta = \sqrt{2} - 1, \csc \theta = -\frac{2}{\sqrt{2-\sqrt{2}}}, \sec \theta = -\frac{2}{\sqrt{2+\sqrt{2}}}, \cot \theta = \sqrt{2} + 1$
7. 2
8. $\sqrt{2}$
9. $\frac{\sqrt{2-\sqrt{2}}}{2}$
10. $-\frac{\sqrt{2+\sqrt{3}}}{2}$
11. $\frac{\sqrt{2-\sqrt{3}}}{2}$
12. $-\frac{\sqrt{2+\sqrt{2}}}{2}$
13. $2 + \sqrt{3}$
14. $-1 - \sqrt{2}$
15. $\sin\left(\frac{x}{2}\right) = \frac{\sqrt{5}}{5}, \cos\left(\frac{x}{2}\right) = -\frac{2\sqrt{5}}{5}, \tan\left(\frac{x}{2}\right) = -\frac{1}{2}$
16. $\sin\left(\frac{x}{2}\right) = \frac{3\sqrt{13}}{13}, \cos\left(\frac{x}{2}\right) = -\frac{2\sqrt{13}}{13}, \tan\left(\frac{x}{2}\right) = -\frac{3}{2}$
17. $\sin\left(\frac{x}{2}\right) = \sqrt{\frac{7+4\sqrt{3}}{14}}, \cos\left(\frac{x}{2}\right) = -\sqrt{\frac{7-4\sqrt{3}}{14}}, \tan\left(\frac{x}{2}\right) = 7 + 4\sqrt{3}$
18. $\sin\left(\frac{x}{2}\right) = \frac{\sqrt{10}}{4}, \cos\left(\frac{x}{2}\right) = \frac{\sqrt{6}}{4}, \tan\left(\frac{x}{2}\right) = \frac{\sqrt{15}}{3}$
19. $\sin(2\theta) = \frac{120}{169}, \cos(2\theta) = \frac{119}{169}, \tan(2\theta) = \frac{120}{119}$
20. $\sin(2\alpha) = \frac{120}{169}, \cos(2\alpha) = -\frac{119}{169}, \tan(2\alpha) = -\frac{120}{119}$
21. $\sin\left(\frac{\theta}{2}\right) = \frac{\sqrt{26}}{26}, \cos\left(\frac{\theta}{2}\right) = \frac{5\sqrt{26}}{26}, \tan\left(\frac{\theta}{2}\right) = \frac{1}{5}$
22. $\sin\left(\frac{\alpha}{2}\right) = \frac{2\sqrt{13}}{13}, \cos\left(\frac{\alpha}{2}\right) = \frac{3\sqrt{13}}{13}, \tan\left(\frac{\alpha}{2}\right) = \frac{2}{3}$
23. $\cos(56^\circ)$
24. $\cos(74^\circ)$
25. $\cos(34^\circ)$
26. $\cos(18x)$
27. $2 \sin(16x)$
28. $3 \sin(10x)$
29. **For exercises 29 – 32, no answers are shown because the solutions will vary.**
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33. $\frac{1+\cos(10x)}{2}$

$$34. \frac{1+\cos(12x)}{2}$$

$$35. \frac{3+\cos(32x)-4\cos(16x)}{8}$$

$$36. \frac{3+\cos(12x)-4\cos(6x)}{8}$$